

Reverse-check review package — HiggsCharacterisation_gen

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	HiggsCharacterisation/model/HiggsCharacterisation_gen.fr
original model name	HiggsCharacterisation_gen (hidden from the agent)
paper	HiggsCharacterisation/text/1306.6464.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

L0v (:=)

```
(-1/4 (ca kHaa gHaa FS[A,mu,nu] FS[A,mu,nu] + sa kAaa gAaa FS[A,mu,nu] Dual[FS][A,mu,nu]) -  
  ↪ 1/2 (ca kHza gHza FS[Z,mu,nu] FS[A,mu,nu] + sa kAza gAza FS[Z,mu,nu] Dual[FS][A,mu,nu])  
  ↪ - 1/4 (ca kHgg gHgg FS[G,mu,nu,a] FS[G,mu,nu,a] + sa kAgg gAgg FS[G,mu,nu,a]  
  ↪ Dual[FS][G,mu,nu,a]) - 1/4/Lambda (ca kHzz FS[Z,mu,nu] FS[Z,mu,nu] + sa kAzz FS[Z,mu,nu]  
  ↪ Dual[FS][Z,mu,nu]) - 1/2/Lambda (ca kHww FS[Wbar,mu,nu] FS[W,mu,nu] + sa kAww  
  ↪ FS[Wbar,mu,nu] Dual[FS][W,mu,nu]) - 1/Lambda (ca kHda Z[nu] del[FS[A,mu,nu],mu] + ca kHdz  
  ↪ Z[nu] del[FS[Z,mu,nu],mu] + ca (kHdw Wbar[nu] del[FS[W,mu,nu],mu] + HC[kHdw Wbar[nu]  
  ↪ del[FS[W,mu,nu],mu])))) XO
```

L1f (:=)

```
(kqa au uqbar[s,n,i].Ga[mu,s,t].uq[t,n,i] + kqa ad dqbar[s,n,i].Ga[mu,s,t].dq[t,n,i] + kla an  
  ↪ vlbar[s,n].Ga[mu,s,t].vl[t,n] + kla al lbar[s,n].Ga[mu,s,t].l[t,n] - kqb bu  
  ↪ uqbar[s,n,i].Ga[mu,s,t].Ga[5,t,u].uq[u,n,i] - kqb bd  
  ↪ dqbar[s,n,i].Ga[mu,s,t].Ga[5,t,u].dq[u,n,i] - klb bn  
  ↪ vlbar[s,n].Ga[mu,s,t].Ga[5,t,u].vl[u,n] - klb bl lbar[s,n].Ga[mu,s,t].Ga[5,t,u].l[u,n])  
  ↪ X1[mu]
```

L1w (:=)

```
I kw1 gwwz (FS[Wbar,mu,nu] W[mu] - FS[W,mu,nu] Wbar[mu]) X1[nu] + I kw2 gwwz Wbar[mu] W[nu]  
  ↪ FS[X1,mu,nu] - kw3 Wbar[mu] W[nu] (del[X1[nu],mu] + del[X1[mu],nu]) + I kw4 Wbar[mu]  
  ↪ W[nu] Dual[FS][X1,mu,nu] - kw5 Eps[mu,nu,rho,sig] (Wbar[mu] del[W[nu],rho] -  
  ↪ del[Wbar[mu],rho] W[nu]) X1[sig]
```

L1z (:=)

```
-kz1 FS[Z,mu,nu] Z[mu] X1[nu] - kz3 X1[mu] del[Z[mu],nu] Z[nu] - kz5 Eps[mu,nu,rho,sig]  
  ↪ X1[mu] Z[nu] del[Z[sig],rho]
```

L1 (:=)

L1f + L1w + L1z

L2f (:=)

$-1/\text{Lambda}$ (kq TFq[mu,nu] + kq3 (TFq3[mu,nu] + TYq[mu,nu]) + kl (TF1[mu,nu] + TY1[mu,nu]))
 $\hookrightarrow$ X2[mu,nu]

L2v (:=)

$-1/\text{Lambda}$ (kg TGg[mu,nu] + ka TGa[mu,nu] + kz TGz[mu,nu] + kw TGw[mu,nu]) X2[mu,nu]

L2 (:=)

L2f + L2v

LHCNP (:=)

L0f + L0v + L1 + L2

Blank-slate reconstruction

Reconstruction of `sanitized.fr`

Conventions

The new fields are denoted

$$X_0, \quad X_{1\mu}, \quad X_{2\mu\nu}.$$

The model uses physical SM gauge fields $A_\mu, Z_\mu, W_\mu^+, W_\mu^-, G_\mu^a$. I write

$$F_{\mu\nu}^V = \partial_\mu V_\nu - \partial_\nu V_\mu$$

for abelian/physical field strengths as they appear in the file, and

$$\tilde{F}_{\mu\nu}^V = \frac{1}{2} \epsilon_{\mu\nu\rho\sigma} F^{V\rho\sigma}.$$

For gluons,

$$G_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc} G_\mu^b G_\nu^c.$$

The file does not declare gauge groups or covariant derivatives directly. It relies on SM symbols and FeynRules field strengths. Where the spin-2 fermion tensors require a covariant derivative, the SM gauge content is the usual one:

$$D_\mu = \partial_\mu + ig_s G_\mu^a T^a + ie Q_f A_\mu + i \frac{e}{s_w c_w} (T_f^3 - Q_f s_w^2) Z_\mu + i \frac{e}{\sqrt{2} s_w} (T^+ W_\mu^+ + T^- W_\mu^-),$$

with the $SU(3)$ term only for quarks and the charged-current $W^\pm$ terms only for left-handed weak doublets.

Lagrangian

The total new-physics Lagrangian is

$$\mathcal{L}_{\text{HCNP}} = \mathcal{L}_{0f} + \mathcal{L}_{0v} + \mathcal{L}_1 + \mathcal{L}_{2f} + \mathcal{L}_{2v}.$$

Scalar X_0 fermion couplings

L0f

$$\mathcal{L}_{0f} = -X_0 c_\alpha \left[k_{Htt} \frac{m_t}{v} \bar{t}t + k_{Hbb} \frac{m_b}{v} \bar{b}b + k_{H\ell\ell} \frac{m_\tau}{v} \bar{\tau}\tau \right].$$

L0f

$$\mathcal{L}_{0f} \supset -iX_0 s_\alpha \left[k_{Att} \frac{m_t}{v} \bar{t}\gamma^5 t + k_{Abb} \frac{m_b}{v} \bar{b}\gamma^5 b + k_{A\ell\ell} \frac{m_\tau}{v} \bar{\tau}\gamma^5 \tau \right].$$

Here $c_\alpha = \mathbf{ca}$, $s_\alpha = \mathbf{sa} = \sqrt{1 - c_\alpha^2}$, and $v = \mathbf{vev}$.

Scalar X_0 gauge-boson couplings

L0v

$$\mathcal{L}_{0v} \supset -\frac{1}{4}X_0 c_\alpha k_{Haa} g_{Haa} F_{\mu\nu}^A F^{A\mu\nu}.$$

L0v

$$\mathcal{L}_{0v} \supset -\frac{1}{4}X_0 s_\alpha k_{Aaa} g_{Aaa} F_{\mu\nu}^A \widetilde{F}^{A\mu\nu}.$$

L0v

$$\mathcal{L}_{0v} \supset -\frac{1}{2}X_0 c_\alpha k_{Hza} g_{Hza} F_{\mu\nu}^Z F^{A\mu\nu}.$$

L0v

$$\mathcal{L}_{0v} \supset -\frac{1}{2}X_0 s_\alpha k_{Aza} g_{Aza} F_{\mu\nu}^Z \widetilde{F}^{A\mu\nu}.$$

L0v

$$\mathcal{L}_{0v} \supset -\frac{1}{4}X_0 c_\alpha k_{Hgg} g_{Hgg} G_{\mu\nu}^a G^{a\mu\nu}.$$

L0v

$$\mathcal{L}_{0v} \supset -\frac{1}{4}X_0 s_\alpha k_{Agg} g_{Agg} G_{\mu\nu}^a \widetilde{G}^{a\mu\nu}.$$

L0v

$$\mathcal{L}_{0v} \supset -\frac{1}{4\Lambda}X_0 c_\alpha k_{Hzz} F_{\mu\nu}^Z F^{Z\mu\nu}.$$

L0v

$$\mathcal{L}_{0v} \supset -\frac{1}{4\Lambda}X_0 s_\alpha k_{Azz} F_{\mu\nu}^Z \widetilde{F}^{Z\mu\nu}.$$

L0v

$$\mathcal{L}_{0v} \supset -\frac{1}{2\Lambda}X_0 c_\alpha k_{Hww} F_{\mu\nu}^{W^-} F^{W^+\mu\nu}.$$

L0v

$$\mathcal{L}_{0v} \supset -\frac{1}{2\Lambda}X_0 s_\alpha k_{Aww} F_{\mu\nu}^{W^-} \widetilde{F}^{W^+\mu\nu}.$$

L0v

$$\mathcal{L}_{0v} \supset -\frac{c_\alpha k_{Hda}}{\Lambda}X_0 Z_\nu \partial_\mu F^{A\mu\nu}.$$

L0v

$$\mathcal{L}_{0v} \supset -\frac{c_\alpha k_{Hdz}}{\Lambda}X_0 Z_\nu \partial_\mu F^{Z\mu\nu}.$$

L0v

$$\mathcal{L}_{0v} \supset -\frac{c_\alpha}{\Lambda}X_0 [k_{Hdw} W_\nu^- \partial_\mu F^{W^+\mu\nu} + k_{Hdw}^* W_\nu^+ \partial_\mu F^{W^-\mu\nu}],$$

with

$$k_{Hdw} = k_{HdwR} + ik_{HdwI}.$$

Vector X_1 fermion couplings

L1f

$$\mathcal{L}_{1f} = X_{1\mu} \sum_{n,i} \bar{u}_{ni} \gamma^\mu (k_{qa} a_u - k_{qb} b_u \gamma^5) u_{ni}.$$

L1f

$$\mathcal{L}_{1f} \supset X_{1\mu} \sum_{n,i} \bar{d}_{ni} \gamma^\mu (k_{qa} a_d - k_{qb} b_d \gamma^5) d_{ni}.$$

L1f

$$\mathcal{L}_{1f} \supset X_{1\mu} \sum_n \bar{\nu}_n \gamma^\mu (k_{\ell a} a_\nu - k_{\ell b} b_\nu \gamma^5) \nu_n.$$

L1f

$$\mathcal{L}_{1f} \supset X_{1\mu} \sum_n \bar{\ell}_n \gamma^\mu (k_{\ell a} a_\ell - k_{\ell b} b_\ell \gamma^5) \ell_n.$$

The internal SM-like vector/axial coefficients are

$$\begin{aligned} a_u &= \frac{e}{2s_w c_w} \left(\frac{1}{2} - \frac{4}{3} s_w^2 \right), & b_u &= \frac{e}{2s_w c_w} \frac{1}{2}, \\ a_d &= \frac{e}{2s_w c_w} \left(-\frac{1}{2} + \frac{2}{3} s_w^2 \right), & b_d &= \frac{e}{2s_w c_w} \left(-\frac{1}{2} \right), \\ a_\nu &= b_\nu = \frac{e}{2s_w c_w} \frac{1}{2}, \\ a_\ell &= \frac{e}{2s_w c_w} \left(-\frac{1}{2} + 2s_w^2 \right), & b_\ell &= \frac{e}{2s_w c_w} \left(-\frac{1}{2} \right). \end{aligned}$$

Vector X_1 charged-vector couplings

L1w

$$\mathcal{L}_{1w} \supset i k_{w1} g_{WWZ} [F_{\mu\nu}^{W^-} W^{+\mu} - F_{\mu\nu}^{W^+} W^{-\mu}] X_1^\nu.$$

L1w

$$\mathcal{L}_{1w} \supset i k_{w2} g_{WWZ} W_\mu^- W_\nu^+ F^{X_1 \mu\nu}.$$

L1w

$$\mathcal{L}_{1w} \supset -k_{w3} W_\mu^- W_\nu^+ (\partial^\mu X_1^\nu + \partial^\nu X_1^\mu).$$

L1w

$$\mathcal{L}_{1w} \supset i k_{w4} W_\mu^- W_\nu^+ \tilde{F}^{X_1 \mu\nu}.$$

L1w

$$\mathcal{L}_{1w} \supset -k_{w5} \epsilon_{\mu\nu\rho\sigma} [W^{-\mu} \partial^\rho W^{+\nu} - (\partial^\rho W^{-\mu}) W^{+\nu}] X_1^\sigma.$$

where

$$g_{WWZ} = -\frac{e c_w}{s_w}.$$

Vector X_1 neutral-vector couplings

L1z

$$\mathcal{L}_{1z} \supset -k_{z1} F_{\mu\nu}^Z Z^\mu X_1^\nu.$$

L1z

$$\mathcal{L}_{1z} \supset -k_{z3} X_{1\mu} (\partial_\nu Z^\mu) Z^\nu.$$

L1z

$$\mathcal{L}_{1z} \supset -k_{z5} \epsilon_{\mu\nu\rho\sigma} X_1^\mu Z^\nu \partial^\rho Z^\sigma.$$

Spin-2 X_2 fermion couplings

L2f

$$\mathcal{L}_{2f} = -\frac{1}{\Lambda} X_2^{\mu\nu} \left[k_q T_{\mu\nu}^q + k_{q3} (T_{\mu\nu}^{q3} + Y_{\mu\nu}^q) + k_\ell (T_{\mu\nu}^\ell + Y_{\mu\nu}^\ell) \right].$$

The file defines

$$T_{\mu\nu}^q = \sum_{f=u,d,c,s} T_{\mu\nu}^f, \quad T_{\mu\nu}^{q3} = T_{\mu\nu}^t + T_{\mu\nu}^b,$$

$$T_{\mu\nu}^\ell = \sum_{f=\nu_e, \nu_\mu, \nu_\tau, e, \mu, \tau} T_{\mu\nu}^f.$$

The fermion energy-momentum tensor represented by `Tfermion[f,mu,nu]` is the symmetric kinetic tensor

$$T_{\mu\nu}^f = \frac{i}{4} [\bar{f} \gamma_\mu D_\nu f - (D_\nu \bar{f}) \gamma_\mu f + \bar{f} \gamma_\nu D_\mu f - (D_\mu \bar{f}) \gamma_\nu f].$$

The extra Yukawa/mass-trace pieces encoded in `TYq` and `TYl` are

$$Y_{\mu\nu}^q = \eta_{\mu\nu} (m_t \bar{t} t + m_b \bar{b} b),$$

$$Y_{\mu\nu}^\ell = \eta_{\mu\nu} m_\tau \bar{\tau} \tau.$$

Spin-2 X_2 gauge-boson couplings

L2v

$$\mathcal{L}_{2v} = -\frac{1}{\Lambda} X_2^{\mu\nu} \left[k_g T_{\mu\nu}^g + k_a T_{\mu\nu}^\gamma + k_z T_{\mu\nu}^Z + k_w T_{\mu\nu}^W \right].$$

The gluon tensor is

$$T_{\mu\nu}^g = \frac{1}{4} \eta_{\mu\nu} G_{\rho\sigma}^a G^{a\rho\sigma} - G_{\mu\rho}^a G_\nu^{a\rho}.$$

The photon tensor is

$$T_{\mu\nu}^\gamma = \frac{1}{4} \eta_{\mu\nu} F_{\rho\sigma}^A F^{A\rho\sigma} - F_{\mu\rho}^A F_\nu^{A\rho}.$$

The Z -boson tensor is

$$T_{\mu\nu}^Z = \eta_{\mu\nu} \left[\frac{1}{4} F_{\rho\sigma}^Z F^{Z\rho\sigma} - \frac{1}{2} m_Z^2 Z_\rho Z^\rho \right] - F_{\mu\rho}^Z F_\nu^{Z\rho} + m_Z^2 Z_\mu Z_\nu.$$

The charged- W tensor is

$$T_{\mu\nu}^W = \eta_{\mu\nu} \left[\frac{1}{2} F_{\rho\sigma}^{W^-} F^{W^+\rho\sigma} - m_W^2 W_\rho^- W^{+\rho} \right] - F_{\mu\rho}^{W^-} F_\nu^{W^+\rho} - F_{\nu\rho}^{W^-} F_\mu^{W^+\rho} + m_W^2 (W_\mu^- W_\nu^+ + W_\nu^- W_\mu^+).$$

Field table

.fr class	Field	Spin	SU(3) rep	SU(2) rep	Q	Y	Self- conjugate	Mass
S[100]	X0	0	singlet, no color index declared	singlet/not declared	0	0	yes	MX0 = 125.0
V[101]	X1	1	singlet, no color index declared	singlet/not declared	0	0	yes	MX1 = 125.0

.fr class	Field	Spin	SU(3) rep	SU(2) rep	Q	Y	Self- conjugate	Mass
T[102]	X2	2	singlet, no color index declared	singlet/not declared	0	0	yes	MX2 = 125.0

Widths are also declared:

$$WX0 = WX1 = WX2 = 0.00407.$$

External parameters

Parameter	Value	Multiplies	Physical meaning
Lambda	1000.	inverse powers in L0v, L2f, L2v	EFT scale suppressing dimension-5 operators
ca	1.	CP-even scalar pieces	scalar CP-even mixing coefficient c_α
kSM	1.	no term in LHCNP in this file	declared external normalization, unused in the shown Lagrangian
kHtt	1.	$X_0 \bar{t} t$	CP-even top Yukawa modifier
kAtt	1.	$X_0 \bar{t} i \gamma^5 t$	CP-odd top Yukawa modifier
kHbb	1.	$X_0 \bar{b} b$	CP-even bottom Yukawa modifier
kAbb	1.	$X_0 \bar{b} i \gamma^5 b$	CP-odd bottom Yukawa modifier
kHll	1.	$X_0 \bar{\tau} \tau$	CP-even tau Yukawa modifier
kAll	1.	$X_0 \bar{\tau} i \gamma^5 \tau$	CP-odd tau Yukawa modifier
kHaa	1.	$X_0 F_{\mu\nu}^A F^{A\mu\nu}$	CP-even $X_0 \gamma\gamma$ coupling modifier
kAaa	1.	$X_0 F_{\mu\nu}^A \tilde{F}^{A\mu\nu}$	CP-odd $X_0 \gamma\gamma$ coupling modifier
kHza	1.	$X_0 F_{\mu\nu}^Z F^{A\mu\nu}$	CP-even $X_0 Z\gamma$ coupling modifier
kAza	1.	$X_0 F_{\mu\nu}^Z \tilde{F}^{A\mu\nu}$	CP-odd $X_0 Z\gamma$ coupling modifier
kHgg	1.	$X_0 G_{\mu\nu}^a G^{a\mu\nu}$	CP-even $X_0 gg$ coupling modifier
kAgg	1.	$X_0 G_{\mu\nu}^a \tilde{G}^{a\mu\nu}$	CP-odd $X_0 gg$ coupling modifier
kHzz	0.	$X_0 F_{\mu\nu}^Z F^{Z\mu\nu} / \Lambda$	CP-even $X_0 ZZ$ EFT coupling
kAzz	0.	$X_0 F_{\mu\nu}^Z \tilde{F}^{Z\mu\nu} / \Lambda$	CP-odd $X_0 ZZ$ EFT coupling
kHww	0.	$X_0 F_{\mu\nu}^{W^-} F^{W^+\mu\nu} / \Lambda$	CP-even $X_0 W^+ W^-$ EFT coupling

Parameter	Value	Multiplies	Physical meaning
kAww	0.	$X_0 F_{\mu\nu}^{W^-} \tilde{F}^{W+\mu\nu} / \Lambda$	CP-odd $X_0 W^+ W^-$ EFT coupling
kHda	0.	$X_0 Z_\nu \partial_\mu F^{A\mu\nu} / \Lambda$	derivative $X_0 Z\gamma$ coupling
kHdz	0.	$X_0 Z_\nu \partial_\mu F^{Z\mu\nu} / \Lambda$	derivative $X_0 ZZ$ coupling
kHdwR	0.	real part of kHdw	real part of derivative $X_0 W^+ W^-$ coupling
kHdwI	0.	imaginary part of kHdw	imaginary part of derivative $X_0 W^+ W^-$ coupling
kqa	1.	vector quark current coupled to X_1	quark vector-current strength
kqb	1.	axial quark current coupled to X_1	quark axial-current strength
kla	1.	vector lepton/neutrino current coupled to X_1	lepton vector-current strength
klb	1.	axial lepton/neutrino current coupled to X_1	lepton axial-current strength
kw1	1.	first $X_1 W^+ W^-$ tensor structure	anomalous charged-vector coupling
kw2	1.	$W^- W^+ F^{X_1}$	anomalous charged-vector coupling
kw3	0.	$W^- W^+ \partial X_1$ symmetric derivative	anomalous charged-vector coupling
kw4	0.	$W^- W^+ \tilde{F}^{X_1}$	CP-odd/dual charged-vector coupling
kw5	0.	$\epsilon W \partial W X_1$	parity-odd charged-vector coupling
kz1	0.	$F^Z Z X_1$	anomalous $X_1 ZZ$ coupling
kz3	1.	$X_1 (\partial Z) Z$	anomalous $X_1 ZZ$ derivative coupling
kz5	0.	$\epsilon X_1 Z \partial Z$	parity-odd $X_1 ZZ$ coupling
kq	1.	$X_2 T_{\mu\nu}$ of light quarks u, d, c, s	spin-2 coupling to light-quark energy-momentum tensor
kq3	1.	$X_2 T_{\mu\nu}$ of t, b plus mass trace	spin-2 coupling to third-generation quark tensor
kl	1.	$X_2 T_{\mu\nu}$ of leptons plus tau mass trace	spin-2 coupling to lepton tensor
kg	1.	$X_2 T_{\mu\nu}^g$	spin-2 coupling to gluon tensor
ka	1.	$X_2 T_{\mu\nu}^\gamma$	spin-2 coupling to photon tensor
kz	1.	$X_2 T_{\mu\nu}^Z$	spin-2 coupling to Z -boson tensor
kw	1.	$X_2 T_{\mu\nu}^W$	spin-2 coupling to charged- W tensor

Internal parameters appearing in the Lagrangian are

$$s_\alpha = \sqrt{1 - c_\alpha^2}, \quad k_{Hdw} = k_{HdwR} + ik_{HdwI},$$

and the loop-normalized effective coefficients

$$g_{Haa}, g_{Aaa}, g_{Hza}, g_{Aza}, g_{Hgg}, g_{Agg}.$$

Physics summary

The file encodes a neutral spin-0, spin-1, and spin-2 resonance sector coupled to SM fermions and electroweak/gluon gauge bosons through Yukawa-like, anomalous gauge-field-strength, vector-current, and energy-momentum-tensor operators. The scalar X_0 has both CP-even and CP-odd couplings controlled by c_α and s_α ; the vector X_1 couples to SM vector/axial currents and to WW/ZZ ; the tensor X_2 couples universally or non-universally to SM stress tensors. These interactions mediate production and decay channels such as $gg, \gamma\gamma, Z\gamma, ZZ, W^+W^-, f\bar{f} \leftrightarrow X_i$, with spin-dependent angular structures.

Paper cross-check

Comparison of `reconstruction.md` Against the Paper Model

Paper Locations

The model is defined in **Section 2, “Effective lagrangian”**. The paper states

$$\mathcal{L}_{\text{HC},J} = \mathcal{L}_{\text{SM}-H} + \mathcal{L}_J$$

in **eq. (2.1)**, where $\mathcal{L}_J$ contains the kinetic and interaction terms of the new bosonic state $X(J^P)$.

Relevant definitions:

- **Spin/parity content:** Section 2, before eq. (2.1): $J^P = 0^+, 0^-, 1^+, 1^-, 2^+$.
- **Spin-0 fermion Lagrangian:** **eq. (2.2)**, with c_α, s_α defined in **eq. (2.3)**.
- **Spin-0 vector Lagrangian:** **eq. (2.4)**.
- **Field strengths and duals:** **eqs. (2.5)-(2.7)**.
- **Spin-1 fermion Lagrangian:** **eq. (2.8)**; quark vector/axial coefficients in **eqs. (2.9)-(2.10)**, with analogous lepton coefficients stated in prose.
- **Spin-1 WW Lagrangian:** **eq. (2.11)**.
- **Spin-1 ZZ Lagrangian:** **eq. (2.12)**.
- **Spin-1 parity restrictions:** **eqs. (2.13)-(2.14)**.
- **Spin-2 fermion Lagrangian:** **eq. (2.15)**.
- **Spin-2 vector Lagrangian:** **eq. (2.16)**.
- **Explicit QED fermion/photon energy-momentum tensors:** **eqs. (2.17)-(2.18)**.
- **Universal RS-like spin-2 limit:** **eq. (2.19)**.
- **Non-universal spin-2 quark/gluon example:** Section 4.1, **eq. (4.1)**.

Term-by-Term Comparison

paper term (eq. ref)	reconstruction term	verdict	notes
$\mathcal{L}_{\text{HC},J} = \mathcal{L}_{\text{SM}-H} + \mathcal{L}_J$, one chosen J sector, eq. (2.1)	$\mathcal{L}_{\text{HCNP}} =$ $\mathcal{L}_{0f} + \mathcal{L}_{0v} + \mathcal{L}_1 + \mathcal{L}_{2f} + \mathcal{L}_{2v}$	disagree	The paper presents a model for a new state $X(J^P)$ with a chosen spin/parity sector; the reconstruction sums spin-0, spin-1, and spin-2 sectors simultaneously. This may reflect implementation packaging, but it is not how eq. (2.1) is written.
New bosonic state $X(J^P)$, $J^P = 0^+, 0^-, 1^+, 1^-, 2^+$, Section 2	Separate $X_0, X_{1\mu}, X_{2\mu\nu}$ fields	agree	Physics content matches the spin assignments, though the paper phrases this as alternative hypotheses rather than necessarily simultaneous particles.
Below-EWSB physical-field EFT; no required $SU(2)_L \times U(1)_Y$ assignment for spin-1, Section 2.2	X_0, X_1, X_2 listed as neutral color/electroweak singlets	agree	For spin-2 the paper explicitly says color, weak, and electromagnetic singlet before eq. (2.15). For spin-1 the paper explicitly avoids specifying EW representation; neutral physical X_1 is consistent with the written ZZ/WW interactions.
Spin-0 Yukawa: $-\sum_{f=t,b,\tau} \bar{\psi}_f (c_\alpha \kappa_{Hff} g_{Hff} + k_{Hbb} m_b/v \bar{b}b +$ $i s_\alpha \kappa_{Aff} g_{Aff} \gamma_5) \psi_f X_0$, eq. (2.2)	$-X_0 c_\alpha [k_{Htt} m_t/v \bar{t}t +$ $k_{H\ell\ell} m_\tau/v \bar{\tau}\tau]$ plus $-i X_0 s_\alpha [\dots \bar{f} \gamma^5 f]$	agree	Same third-generation scalar and pseudoscalar Yukawa structures, with $g_{Hff} = g_{Aff} = m_f/v$.
$c_\alpha = \cos \alpha$, $s_\alpha = \sin \alpha$, eq. (2.3)	$c_\alpha = \mathbf{ca}$, $s_\alpha = \sqrt{1 - c_\alpha^2}$	agree	Equivalent only if the implementation restricts to the principal positive s_α . The paper treats s_α as $\sin \alpha$.
SM-like spin-0 vector mass terms $c_\alpha \kappa_{\text{SM}} [\frac{1}{2} g_{HZZ} Z_\mu Z^\mu +$ $g_{HWW} W_\mu^+ W^{-\mu}] X_0$, eq. (2.4)	kSM declared but “unused”; no $X_0 ZZ$ or $X_0 W^+ W^-$ mass term listed	missing-in-reconstruction	This is a central term in eq. (2.4), needed to recover the SM limit with $c_\alpha = 1, \kappa_{\text{SM}} = 1$.
$-\frac{1}{4} [c_\alpha \kappa_{H\gamma\gamma} g_{H\gamma\gamma} A_{\mu\nu} A^{\mu\nu} +$ $s_\alpha \kappa_{A\gamma\gamma} g_{A\gamma\gamma} A_{\mu\nu} \tilde{A}^{\mu\nu}] X_0$, eq. (2.4)	$-\frac{1}{4} X_0 c_\alpha k_{Haa} g_{Haa} F^A F^A,$ $-\frac{1}{4} X_0 s_\alpha k_{Aaa} g_{Aaa} F^A \tilde{F}^A$	agree	Same CP-even and CP-odd $\gamma\gamma$ structures.
$-\frac{1}{2} [c_\alpha \kappa_{HZ\gamma} g_{HZ\gamma} Z_{\mu\nu} A^{\mu\nu} +$ $s_\alpha \kappa_{AZ\gamma} g_{AZ\gamma} Z_{\mu\nu} \tilde{A}^{\mu\nu}] X_0$, eq. (2.4)	$-\frac{1}{2} X_0 c_\alpha k_{Hza} g_{Hza} F^Z F^A,$ $-\frac{1}{2} X_0 s_\alpha k_{Aza} g_{Aza} F^Z \tilde{F}^A$	agree	Same $Z\gamma$ field-strength structures.
$-\frac{1}{4} [c_\alpha \kappa_{Hgg} g_{Hgg} G_{\mu\nu}^a G^{a\mu\nu} +$ $s_\alpha \kappa_{Agg} g_{Agg} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}] X_0$, eq. (2.4)	$-\frac{1}{4} X_0 c_\alpha k_{Hgg} g_{Hgg} GG,$ $-\frac{1}{4} X_0 s_\alpha k_{Agg} g_{Agg} G\tilde{G}$	agree	Same gluonic CP-even and CP-odd structures.

paper term (eq. ref)	reconstruction term	verdict	notes
$-\frac{1}{4\Lambda}[c_\alpha\kappa_{HZZ}Z_{\mu\nu}Z^{\mu\nu} + s_\alpha\kappa_{AZZ}Z_{\mu\nu}\tilde{Z}^{\mu\nu}]X_0$, eq. (2.4)	$-\frac{1}{4\Lambda}X_0c_\alpha k_{Hzz}F^Z F^Z$, $-\frac{1}{4\Lambda}X_0s_\alpha k_{Azz}F^Z \tilde{F}^Z$	agree	Same higher-dimensional ZZ field-strength terms.
$-\frac{1}{2\Lambda}[c_\alpha\kappa_{HWW}W_{\mu\nu}^+W^{-\mu\nu} + s_\alpha\kappa_{AWW}W_{\mu\nu}^+\tilde{W}^{-\mu\nu}]X_0$, eq. (2.4)	$-\frac{1}{2\Lambda}X_0c_\alpha k_{Hww}F^{W-}F^{W+}$, $-\frac{1}{2\Lambda}X_0s_\alpha k_{Aww}F^{W-}\tilde{F}^{W+}$	agree	Same W^+W^- field-strength content; charge ordering is immaterial for these bilinears.
$-\frac{c_\alpha}{\Lambda}[\kappa_{H\partial\gamma}Z_\nu\partial_\mu A^{\mu\nu} + \kappa_{H\partial Z}Z_\nu\partial_\mu Z^{\mu\nu} + \kappa_{H\partial W}W_\nu^+\partial_\mu W^{-\mu\nu} + \text{h.c.}]X_0$, eq. (2.4)	$-\frac{c_\alpha}{\Lambda}X_0[k_{Hda}Z_\nu\partial F_A + k_{Hdz}Z_\nu\partial F_Z + k_{Hdw}W_\nu^-\partial F_{W^+} + k_{Hdw}^*W_\nu^+\partial F_{W^-}]$	disagree	Neutral derivative terms agree. The charged W term uses the complex coefficient on the charge-conjugate structure relative to the paper, so the imaginary part has the opposite convention unless k_{Hdw} is defined as $\kappa_{H\partial W}^*$. Matches the paper definitions.
Field strengths $V_{\mu\nu} = \partial_\mu V_\nu - \partial_\nu V_\mu$, $G_{\mu\nu}^a = \partial G + g_s f^{abc}G^b G^c$, dual $\tilde{V}_{\mu\nu} = \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}V^{\rho\sigma}$, eqs. (2.5)-(2.7)	Same definitions for physical $A, Z, W^\pm$, gluons, and duals	agree	
Spin-1 fermions $\sum_{f=q,\ell}\bar{\psi}_f\gamma^\mu(\kappa_{fa}a_f - \kappa_{fb}b_f\gamma^5)\psi_f X_{1\mu}$, eq. (2.8)	Separate sums over u, d, ν, ℓ with $k_{qa}, k_{qb}, k_{\ell a}, k_{\ell b}$ and SM-like a_f, b_f	agree	Same vector/axial current structure; reconstruction makes generation and color sums explicit.
Quark coefficients a_u, b_u, a_d, b_d , eqs. (2.9)-(2.10)	Same coefficients written with $e/(s_w c_w)$	agree	Since $g = e/s_w$, the expressions match.
Lepton coefficients “similarly for the leptons,” after eq. (2.10)	Explicit $a_\nu = b_\nu = e/(4s_w c_w)$, $a_\ell = e(-1/2 + 2s_w^2)/(2s_w c_w)$, $b_\ell = -e/(4s_w c_w)$	agree	These are the standard SM Z -like vector/axial coefficients implied by the paper.
$i\kappa_{W1}g_{WWZ}(W_{\mu\nu}^+W^{-\mu} - W_{\mu\nu}^-W^{+\mu})X_1^\nu$, eq. (2.11)	$ik_{w1}g_{WWZ}(F_{\mu\nu}^{W-}W^{+\mu} - F_{\mu\nu}^{W+}W^{-\mu})X_1^\nu$	disagree	This is the negative of the paper’s bracket if $k_{w1} = \kappa_{W1}$ and the same g_{WWZ} convention is used. Because $X_1^{\mu\nu}$ is antisymmetric, exchanging $W^+ \leftrightarrow W^-$ with the Lorentz indices gives an overall minus sign.
$i\kappa_{W2}g_{WWZ}W_\mu^+W_\nu^-X_1^{\mu\nu}$, eq. (2.11)	$ik_{w2}g_{WWZ}W_\mu^-W_\nu^+F_{X_1}^{\mu\nu}$	disagree	The derivative tensor is symmetric in μ, ν , so the charge/index ordering is equivalent.
$-\kappa_{W3}W_\mu^+W_\nu^-(\partial^\mu X_1^\nu + \partial^\nu X_1^\mu)$, eq. (2.11)	$-k_{w3}W_\mu^-W_\nu^+(\partial^\mu X_1^\nu + \partial^\nu X_1^\mu)$	agree	

paper term (eq. ref)	reconstruction term	verdict	notes
$+i\kappa_{W4}W_\mu^+W_\nu^-\tilde{X}_1^{\mu\nu}$, eq. (2.11)	$+ik_{w4}W_\mu^-W_\nu^+\tilde{F}_{X_1}^{\mu\nu}$	disagree	Same antisymmetry issue as the $W2$ term: the reconstruction has the opposite sign for equal coupling definitions.
$-\kappa_{W5}\epsilon_{\mu\nu\rho\sigma}[W^{+\mu}\partial^\rho W^{-\nu} - (\partial^\rho W^{+\mu})W^{-\nu}]X_1^\sigma$, eq. (2.11)	$-k_{w5}\epsilon_{\mu\nu\rho\sigma}[W^{-\mu}\partial^\rho W^{+\nu} - (\partial^\rho W^{-\mu})W^{+\nu}]X_1^\sigma$	agree	After relabeling the antisymmetric $\epsilon_{\mu\nu\rho\sigma}$ indices, this is equivalent.
$g_{WWZ} = -e \cot \theta_W$, below eq. (2.11)	$g_{WWZ} = -ec_w/s_w$	agree	Same definition.
$-\kappa_{Z1}Z_{\mu\nu}Z^\mu X_1^\nu$, eq. (2.12)	$-k_{z1}F_{\mu\nu}^ZZ^\mu X_1^\nu$	agree	Same structure.
$-\kappa_{Z3}X_{1\mu}(\partial_\nu Z^\mu)Z^\nu$, eq. (2.12)	$-k_{z3}X_{1\mu}(\partial_\nu Z^\mu)Z^\nu$	agree	Same structure.
$-\kappa_{Z5}\epsilon_{\mu\nu\rho\sigma}X_1^\mu Z^\nu \partial^\rho Z^\sigma$, eq. (2.12)	Same $k_{z5}\epsilon XZZ$ structure	agree	Same structure.
No effective $\mathcal{L}_1^\gamma$, after eq. (2.12)	No $X_1\gamma\gamma$ or X_1gg term listed	agree	Matches the paper's Landau-Yang discussion for on-shell spin-1.
Parity restrictions for pure 1^- and 1^+ , eqs. (2.13)-(2.14)	General X_1 terms retained with independent coefficients	agree	The reconstruction describes the general implemented interaction set; the paper gives restrictions for pure-parity limits.
Spin-2 fermions $-\frac{1}{\Lambda}\sum_{f=q,\ell}\kappa_f T_{\mu\nu}^f X_2^{\mu\nu}$, eq. (2.15)	$-\frac{1}{\Lambda}X_2^{\mu\nu}[k_q T_{\mu\nu}^q + k_{q3}(T_{\mu\nu}^{q3} + Y_{\mu\nu}^q) + k_\ell(T_{\mu\nu}^\ell + Y_{\mu\nu}^\ell)]$	disagree	The paper's eq. (2.15) uses $\kappa_f T_{\mu\nu}^f$ for quarks/leptons; reconstruction introduces an explicit separate third-generation quark coefficient k_{q3} and separate mass-trace pieces Y^q, Y^ℓ .
Explicit QED fermion $T_{\mu\nu}^f$, eq. (2.17), including $-g_{\mu\nu}[\psi(i\gamma^\rho D_\rho - m_f)\psi - \frac{1}{2}\partial_\rho(\bar{\psi}i\gamma^\rho\psi)]$ and derivative terms	Symmetric kinetic tensor $\frac{i}{4}[\bar{f}\gamma_\mu D_\nu f - (D_\nu \bar{f})\gamma_\mu f + \mu \leftrightarrow \nu]$, plus separate mass traces for t, b, τ	disagree	The reconstruction is not the same off-shell tensor as eq. (2.17); mass and total-derivative/ $g_{\mu\nu}$ pieces are treated differently and only some fermion masses are retained.
Spin-2 vectors $-\frac{1}{\Lambda}\sum_{V=Z,W,\gamma,g}\kappa_V T_{\mu\nu}^V X_2^{\mu\nu}$, eq. (2.16)	$-\frac{1}{\Lambda}X_2^{\mu\nu}[k_g T^g + k_a T^a + k_z T^Z + k_w T^W]_{\mu\nu}$	agree	Same coupling pattern to gluon, photon, Z , and W energy-momentum tensors.

paper term (eq. ref)	reconstruction term	verdict	notes
Explicit photon $T_{\mu\nu}^\gamma$, eq. (2.18), containing $\partial_\rho \partial_\sigma A^\sigma A^\rho$, $(\partial_\rho A^\rho)^2$, and $\partial_\mu \partial_\rho A^\rho A_\nu$ terms	Maxwell form $T_{\mu\nu}^\gamma = \frac{1}{4}\eta_{\mu\nu}F^2 - F_{\mu\rho}F_\nu{}^\rho$	disagree	The reconstruction omits the derivative/gauge-fixing-like terms present in the paper's explicit eq. (2.18). They may vanish under specific gauge/on-shell conditions but are part of the displayed paper tensor.
Spin-2 gluon tensor, implied by eq. (2.16) and standard E-M tensor	$\frac{1}{4}\eta_{\mu\nu}G^2 - G_{\mu\rho}G_\nu{}^\rho$	agree	This matches the standard gauge-field E-M tensor structure, up to metric-sign convention.
Spin-2 massive Z, W tensors, implied by eq. (2.16) and references	Proca-like Z and $W^\pm$ stress tensors with mass terms	agree	The paper does not print these explicitly, but the reconstruction matches the expected massive-vector E-M tensor content.
Universal RS-like limit $\kappa_f = \kappa_V \forall f, V$, eq. (2.19)	Independent $k_q, k_{q3}, k_\ell, k_g, k_a, k_z, k_w$	agree	The reconstruction gives the general non-universal parameterization; the universal limit is obtained by setting all relevant couplings equal, modulo the extra k_{q3} .
Non-universal quark/gluon example $-\kappa_q T^q X_2/\Lambda - \kappa_g T^g X_2/\Lambda$, Section 4.1 eq. (4.1)	Independent k_q and k_g	agree	Matches the paper's non-universal spin-2 discussion for quarks and gluons.

Disagreements and Severity

disagreement	severity	what a human should check
Reconstruction sums spin-0, spin-1, and spin-2 sectors into one $\mathcal{L}_{\text{HCNP}}$, whereas the paper writes $\mathcal{L}_{\text{HC},J}$ for a chosen J sector in eq. (2.1).	convention	Check whether the implementation file intentionally contains all hypotheses simultaneously with couplings used as switches.
Reconstruction omits the SM-like spin-0 $X_0 ZZ$ and $X_0 W^+ W^-$ mass terms proportional to $c_\alpha \kappa_{\text{SM}}$ in eq. (2.4).	substantive	Check whether those terms are present elsewhere in the implementation or whether $\mathbf{kSM}$ being unused is an extraction/reconstruction error.
Charged spin-0 derivative W term uses the complex coefficient on the charge-conjugate structure relative to eq. (2.4).	convention	Check the implementation's definition of $\mathbf{kHdW}$; it may be the complex conjugate of the paper's $\kappa_{H\partial W}$.
Spin-1 $W1$ term has the opposite sign relative to eq. (2.11) if $k_{w1} = \kappa_{W1}$.	substantive	Check the charged- W field naming and sign convention in the implementation, including the definition of $W_{\mu\nu}^\pm$.

disagreement	severity	what a human should check
Spin-1 $W2$ term has the opposite sign relative to eq. (2.11) because the charged fields carry swapped Lorentz indices contracted with antisymmetric $X_1^{\mu\nu}$.	substantive	Check whether the implementation defines <code>kw2</code> with an implicit minus sign relative to the paper.
Spin-1 $W4$ term has the opposite sign relative to eq. (2.11) for the same antisymmetric-index reason as $W2$.	substantive	Check whether the implementation defines <code>kw4</code> with an implicit minus sign relative to the paper.
Spin-2 fermion sector introduces a separate k_{q3} third-generation quark coupling and extra mass-trace pieces not present in the compact paper eq. (2.15).	substantive	Check whether the implementation extends the paper’s displayed κ_q, κ_ℓ structure or whether the paper’s $\sum_f$ notation was intended to allow generation-dependent κ_f .
Reconstruction’s fermion energy-momentum tensor differs from the explicit paper tensor in eq. (2.17), especially in $g_{\mu\nu}$, mass, and total-derivative terms.	substantive	Check the exact <code>FeynRules</code> definition of <code>Tfermion</code> and whether equations of motion or on-shell simplifications were applied in the reconstruction.
Reconstruction’s photon energy-momentum tensor omits derivative terms shown explicitly in paper eq. (2.18).	substantive	Check whether the implementation uses a simplified transverse/on-shell tensor or includes the omitted terms elsewhere through gauge-fixing conventions.
Reconstruction retains only t, b, τ mass-trace pieces in spin-2 fermion terms.	convention	Check whether light-fermion masses are intentionally neglected in the implementation, as is common phenomenologically.

Overall Assessment

The reconstruction captures the broad Higgs Characterisation model structure: a below-EWSB EFT for neutral spin-0, spin-1, and spin-2 resonance hypotheses, with scalar/pseudoscalar Yukawa and field-strength operators, vector/axial spin-1 fermion currents, anomalous $X_1 WW/X_1 ZZ$ interactions, and spin-2 couplings to SM energy-momentum tensors. The main physics-level gaps are the missing SM-like scalar $X_0 ZZ/X_0 WW$ mass terms, several charged $X_1 WW$ sign differences, and a spin-2 energy-momentum tensor reconstruction that is not identical to the explicit tensors printed in the paper. Some differences may be implementation conventions or on-shell simplifications, but they are important enough that a reviewer should inspect the original implementation definitions before treating the reconstruction as a faithful transcription of the paper.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field’s representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 62 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model’s identity.

Physicist sign-off

- Reviewed by: _____ Date: _____
- Verdict: ☐ approve ☐ approve with corrections ☐ reject
- Notes: